

API 610 CENTRIFUGAL PROCESS PUMPS

OEM TECHNICAL SPECIFICATION & MAINTENANCE MANUAL (REV 4.2)

SECTION 8 · CASING INTEGRITY & WEAR TOLERANCES

8.1 Purpose and Scope:

This section prescribes the minimum allowable wall thicknesses, corrosion allowances, and mandatory structural retirement criteria for Model SX-4000 series centrifugal pumps (including Reflux Pump 3B).

8.2 Structural Minimum Wall Thickness Criteria:

All pressure-retaining components are designed in accordance with ASME Section VIII and API 610 11th Edition standards. Continuous operation is predicated on maintaining a structural safety factor ≥ 2.5 against internal burst pressure.

TABLE 8.4 · MANDATORY CASING SHELL REPLACEMENT LIMITS

Equipment Model / Asset	Nominal Baseline	Minimum Allowable (Limit)	Action on Breach
SX-2000 Series (Water Feed)	4.00 mm	3.00 mm	Scheduled Overhaul
SX-4000 Series (Reflux Pump 3B)	4.80 mm	4.00 mm	MANDATORY SHUTDOWN
SX-6000 Series (High Pressure Gas)	6.50 mm	5.20 mm	MANDATORY SHUTDOWN

CRITICAL OPERATIONAL REQUIREMENT (TABLE 8.4 CLAUSE C):

For Pump 3B (Model SX-4000), the absolute minimum allowable shell thickness is 4.00 mm. Operation with wall thickness below 4.00 mm (e.g. ≤ 3.99 mm) voids all manufacturer safety warranties and constitutes an uncontained containment breach hazard under operating reflux pressures. If inspection reveals wall thickness < 4.00 mm or any through-thickness fatigue cracking, the unit MUST BE DE-PRESSURIZED IMMEDIATELY AND REPLACED.